FeedFlipNets: Feedback-Driven Bit-Flips for Ternary Networks, Activation-Routed DFA, and the Per-Weight Sign Barrier to Transport-Free Learning

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Abstract

FeedFlipNets trains ternary networks by using a cheap feedback signal to *flip bits directly*: weights are stored only as ternary values $\{-1, 0, +1\}$ plus a small integer flip-accumulator—*no float shadow weights*—and a per-weight vote flips a bit when enough evidence accrues. Because a flip needs only the *sign* of "should this weight go up or down," the natural feedback is Direct Feedback Alignment (DFA): it broadcasts the output error transport-free, with no backward lock and no shadow storage. We make three contributions. (i) The bit-flip mechanism is genuinely efficient: with an accurate sign it matches float-shadow backpropagation at $\approx 5.6\times$ less state and $\approx 2\times$ faster steps. (ii) To supply a better transport-free sign we introduce **Activation-Routed DFA** (AR-DFA), which reuses the parts of a Transformer block's Jacobian that are already cached in the forward pass—the softmax routing matrix, its Jacobian, the LayerNorm scale, the nonlinearity mask—and surrogates *only* the weight-transposes; this makes the value-path gradients *exact* (matching autograd to $\sim 10^{-15}$) and, with a perturbation-taught surrogate, cuts the worst-case attention-block *cosine*-alignment angle from $\approx 90°$ to $46.4°$, all lock-free. (iii) The central, and negative, finding: this alignment gain does *not* help bit-flipping. On a ternary Transformer, transport-free flipping is worse than not flipping at all, and AR-DFA does not beat vanilla DFA—because per-weight *sign*-match to the true gradient stays near chance ($p \approx 0.53$) even as the *cosine* alignment improves sharply. Aggregate alignment is the wrong proxy: the binding constraint for transport-free discrete/ternary learning is per-weight *sign correctness*. We support this with pre-registered gates across depth, attention, accuracy, per-step cost, an analytical pipeline model, and a transport-fraction frontier; every guard and number is reproducible.

Keywords: Ternary Networks, Bit-Flip Training, Direct Feedback Alignment, Weight Transport, Sign Correctness, Credit Assignment, Attention, Negative Results, Reproducible Research.

Abbreviations: AR-DFA = Activation-Routed DFA; PT = Perturbation-Taught feedback; BP = backpropagation; bpc = bits per character; CI = confidence interval.

1 Introduction

Backpropagation (BP) computes exact gradients but pays for it structurally: the backward pass is a sequential chain that transports the transpose of each weight matrix and cannot begin at a layer until the layer above has finished. *Direct Feedback Alignment* (DFA) [**? ?**] removes weight transport by broadcasting the output error δ_L directly to each hidden layer ℓ through a fixed matrix B_ℓ. This makes every layer's update depend only on the shared broadcast error—the backward "lock" is gone, and updates can proceed in parallel. That property is the entire reason to prefer DFA: it is the one axis on which a transport-free method could plausibly beat BP (parallel, low-communication training on the right hardware).

DFA's difficulty is equally well known: it trades exactness for a random surrogate, and the surrogate degrades where models are largest. The alignment between the DFA update and the true gradient decays with depth [**?** **?**], and it is especially poor on self-attention, whose effective Jacobian depends on the input and therefore *moves every forward pass*, so a single fixed B_ℓ cannot track it [**?**]. The natural question is whether a better—but still transport-free—feedback rule can close enough of this alignment gap to make DFA worth its promise.

This paper answers that question in two parts. First, we show that a large part of the alignment gap on attention is closable transport-free, by observing that a Transformer block's backward pass factors into two kinds of maps. *Weight maps* (the projection transposes $W_Q^\top, W_K^\top, W_V^\top, W_O^\top, W_1^\top, W_2^\top$) are exactly the weight transport DFA forbids. *Data maps*—the softmax routing matrix $A = \mathrm{softmax}(QK^\top/\sqrt{d})$, its Jacobian, the LayerNorm scale, and the nonlinearity mask—are functions of the activations, are *already cached in the forward pass*, and reusing them is therefore *not* weight transport. **Activation-Routed DFA** (AR-DFA) reuses the data maps exactly and approximates only the weight transposes. Two consequences fall out cleanly: the value/output-projection gradients become *exact* at zero transport cost, and—when the residual surrogate is adapted by a transport-free perturbation rule—the worst-case attention-block alignment angle drops from $\approx 90°$ to $46.4°$.

Second, we put the mechanism to work in the setting that motivates the project: **FeedFlip**, a bit-flip training rule for ternary networks. Weights are stored only as ternary values plus a small integer flip-accumulator—no float shadow weights—and each step casts a per-weight vote in the descent direction, flipping a bit once the accumulated evidence crosses a threshold. A flip consumes only the *sign* of the update, so DFA's cheap, transport-free, shadow-free signal is the natural feedback, and the state per weight collapses from a 32-bit float shadow to a few bits. With an accurate sign this mechanism is efficient and effective: it matches float-shadow BP at $\approx 5.6\times$ less state and $\approx 2\times$ faster steps.

Third, and this is the paper's main scientific message, we treat "does the alignment gain help?" as a falsifiable question with pre-registered gates, and report the answer honestly: **it does not.** On a ternary Transformer, transport-free bit-flipping is *worse than not flipping at all*, and AR-DFA does not beat vanilla DFA. The reason is precise and, we think, the useful lesson of the paper: a bit-flip needs the per-weight *sign* of the gradient, and both DFA and AR-DFA match that sign only near chance ($p \approx 0.53$)—*even as* AR-DFA's aggregate *cosine* alignment improves from $90°$ to $46°$. Cosine alignment—the standard DFA metric—is thus the wrong proxy for what discrete, transport-free learning actually needs. We corroborate the same separation on the continuous side (the alignment gain does not make DFA competitive on accuracy, per-step cost, or wall-clock-to-target either), so the binding constraint on transport-free learning is not aggregate alignment but per-weight sign correctness.

Contributions.

- **FeedFlip** (Sec. **??**): a shadow-free, sign-driven bit-flip training rule for ternary networks. Weights are ternary plus a small integer accumulator; with an accurate sign it matches float-shadow BP at $\approx 5.6\times$ less state and $\approx 2\times$ faster steps.
- **Activation-Routed DFA** (Sec. **??**): a transport-free feedback rule that reuses a Transformer block's cached data-maps and surrogates only the weight transposes, yielding *value-path-exact* gradients (Prop. **??**) and a large improvement in attention *cosine* alignment—our best attempt at a good transport-free sign.
- **The per-weight sign barrier** (Secs. **??**–**??**): the central finding. Transport-free flipping fails on a ternary Transformer and AR-DFA does not rescue it, because per-weight sign-match stays

near chance *even as* cosine alignment improves sharply. Cosine alignment is the wrong proxy; per-weight sign correctness is the binding constraint. We corroborate the same separation on continuous training (accuracy, per-step cost, wall-clock-to-target) via pre-registered gates; every guard and number is reproducible.

2 Background and Related Work

Feedback alignment. Feedback alignment [**?**] replaces the transposed forward weights in the backward pass with fixed random matrices; DFA [**?**] broadcasts the output error directly to each hidden layer. Over training the forward weights partially align to the feedback matrices, so the random update acquires a descent component [**?**]. Scaling studies find the approach competitive on shallow networks but increasingly far from BP as depth and task difficulty grow [**?**], with attention-based and convolutional models the hardest cases [**?**].

Learning the feedback. A separate line closes part of the gap by *learning* the feedback to track $W^\top$: weight mirrors and the Kolen–Pollack rule [**?**], and target-projection variants such as DRTP [**?**]. Methods that recover BP essentially exactly—predictive coding and target propagation [**?**]—do so through iterative, layer-local settling that reintroduces a sequential dependency, i.e. the very cost DFA removes. AR-DFA occupies a different point: it keeps the fixed broadcast across layers (preserving the lock-free property) and instead recovers exactness *within* a block by reusing cached data-maps, learning only a small residual surrogate.

Removing the backward lock. Synthetic gradients / decoupled neural interfaces [**?**] and decoupled greedy learning also break update-locking, at the cost of auxiliary models or local objectives. DFA is a zero-extra-parameter point in this space; our cost model (Sec. **??**) borrows the pipeline-bubble and activation-memory accounting from GPipe and 1F1B/PipeDream [**? ?**] to quantify the wall-clock consequences.

Ternary and discrete training. Ternary and binary networks quantize weights to $\{-1, 0, +1\}$ or $\{\pm 1\}$ and are typically trained with a float *shadow* weight and a straight-through estimator [**? ?**]; sign-based optimizers such as signSGD [**?**] show that the gradient *sign* alone can drive learning. FeedFlip combines these observations—discrete weights and sign-only updates—and removes the float shadow entirely, storing only the ternary value and a small flip-accumulator. An earlier version of this project studied deterministic DFA with ternary *shadow*-weight forwards as a stability baseline on small standard benchmarks; that shadow-weight variant is distinct from the shadow-free bit-flip rule here and is summarized in Appendix **??**.

3 Method: Activation-Routed DFA

3.1 Setup and the transport-free invariant

We study a pre-LayerNorm, single-head decoder block. With residual input x,

$$y = \mathrm{LN}(x), \quad Q = yW_Q,\ K = yW_K,\ V = yW_V, \tag{1}$$

$$A = \mathrm{softmax}\big(QK^\top/\sqrt{d}\big), \quad \mathrm{Ctx} = AV, \quad x_1 = x + \mathrm{Ctx}\,W_O, \tag{2}$$

$$x_2 = x_1 + W_2\,\phi(W_1\,\mathrm{LN}(x_1)), \tag{3}$$

with a causal mask applied inside the softmax for language modeling. A stack of such blocks maps token embeddings to logits.

Definition 1 (Lock-free feedback)**.** *A feedback rule is* lock-free *if, inside the computation of any parameter gradient, the only tensors that cross a layer boundary are (i) the shared broadcast error e and (ii) a scalar loss/perturbation signal. In particular it never reads a downstream layer's weight-transpose $W_{\ell+1}^\top$ or per-layer error $\delta_{\ell+1}$.*

Reusing a layer's *own* cached forward tensors is permitted (plain DFA already reuses $\phi'(z_\ell)$). The distinction that matters is between reading a downstream *weight* (transport) and reusing a cached *activation* (not transport). We verify Def. **??** operationally with a two-part probe (Sec. **??**).

3.2 The Jacobian decomposition

A block receives its incoming error as a direct broadcast of the output error; we use one broadcast point *per pre-LN sublayer*, $e_{\text{attn}} \approx \partial\mathcal{L}/\partial x_1$ and $e_{\text{mlp}} \approx \partial\mathcal{L}/\partial x_2$, which is both the lock-free decomposition and what makes the value-path exact (below). Tracing the exact backward through Eqs. (**??**)–(**??**) and tagging each factor:

gradient	factor	status
$\partial\mathcal{L}/\partial W_O$	$\text{Ctx}^\top e_{\text{attn}}$	**exact** (both operands available)
$\partial\mathcal{L}/\partial\text{Ctx}$	$e_{\text{attn}} W_O^\top$	transport $\rightarrow$ surrogate R_O
$\partial\mathcal{L}/\partial V$	$A^\top(\cdot)$	A cached (exact)
$\partial\mathcal{L}/\partial A$	$(\cdot)V^\top$	V cached (exact)
$\partial\mathcal{L}/\partial(QK^\top)$	$J_A(\cdot)$	softmax-Jac from cached A (exact)
$\partial\mathcal{L}/\partial Q, \partial K$	$(\cdot)K$, $(\cdot)^\top Q$	Q, K cached (exact)
$\partial\mathcal{L}/\partial W_{Q,K,V}$	$y^\top(\cdot)$	y cached (exact)

The *only* transport step in the attention sublayer is $\partial\mathcal{L}/\partial\text{Ctx} = e_{\text{attn}} W_O^\top$; AR-DFA replaces $W_O^\top$ with a surrogate R_O and routes everything else through the cached data-maps exactly. The MLP sublayer is identical in form: $\partial\mathcal{L}/\partial W_2 = H^\top e_{\text{mlp}}$ is exact, one surrogate R_2 replaces $W_2^\top$, and the nonlinearity mask ϕ' is reused exactly. The fixed-DFA baseline, by contrast, broadcasts e_{attn} to each of Q, K, V through independent random matrices and *ignores A entirely*; the ablation in Sec. **??** isolates the win to the reuse of A.

Remark 1 (What is exact and what is not)**.** AR-DFA *tracks the input-conditioned* mixing *(A, J_A, LN scale, ϕ') exactly; the input-conditioned* projection *through $W_{\{Q,K,V,O,1,2\}}$ is approximated, because it factors as (fixed weight)×(input) and the weight half is forbidden. The benefit is thus asymmetric: strong on the value/A-contraction path, weaker on the score path (W_Q, W_K), which sits behind the random surrogate.*

3.3 Perturbation-Taught surrogate adaptation

Left fixed and random, R_O bottlenecks the score path. We adapt it *transport-free* by a Kolen–Pollack-style regression toward $\partial\mathcal{L}/\partial\text{Ctx}$, where that target is estimated by antithetic *node perturbation* of Ctx scored by a forward-only recompute of the (linearized) block loss—no autograd, no $W_O^\top$:

$$\hat{g}_{\text{Ctx}} = \frac{\mathcal{L}(\text{Ctx}+\xi) - \mathcal{L}(\text{Ctx}-\xi)}{2\rho^2}\,\xi, \qquad R_O \leftarrow R_O + \eta_R\, e_{\text{attn}}^\top\big(\hat{g}_{\text{Ctx}} - e_{\text{attn}} R_O\big). \tag{4}$$

We refer to Eq. (**??**) as *Perturbation-Taught* feedback (PT). It reads only the broadcast error, a scalar loss difference, and its own state, so it preserves Def. **??**. Its cost, and its ceiling, are analyzed

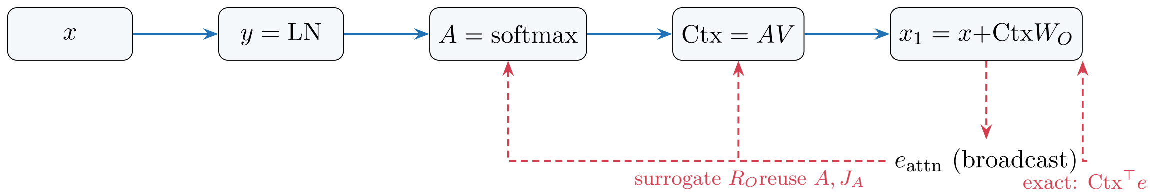

Figure 1: AR-DFA in the attention sublayer. Forward (blue) caches A, Ctx, Q, K, V. The broadcast error e_{attn} (red, dashed) gives the *exact* $\partial\mathcal{L}/\partial W_O = \mathrm{Ctx}^\top e_{\mathrm{attn}}$, is routed through the cached data-maps A, J_A exactly, and passes the single forbidden transpose $W_O^\top$ through the surrogate R_O (adapted by PT). No downstream weight is read.

in Sec. **??**: the estimator variance scales with the perturbed dimension, and the update's fixed point is a fixed linear map.

3.4 Schematic

4 Theory

4.1 Value-path exactness

Proposition 1 (Value-path exactness)**.** *Fix a block and suppose the attention sublayer receives the broadcast error $e_{attn} = \partial\mathcal{L}/\partial x_1$ and the MLP sublayer $e_{mlp} = \partial\mathcal{L}/\partial x_2$. Then the* AR-DFA *gradients for the output projections are exactly the backpropagation gradients, independent of the surrogates R_O, R_2:*

$$\nabla_{W_O}^{\text{AR-DFA}}\mathcal{L} = \mathrm{Ctx}^\top e_{attn} = \nabla_{W_O}^{\text{BP}}\mathcal{L}, \qquad \nabla_{W_2}^{\text{AR-DFA}}\mathcal{L} = H^\top e_{mlp} = \nabla_{W_2}^{\text{BP}}\mathcal{L}. \tag{5}$$

Sketch. $x_1 = x + \mathrm{Ctx}\, W_O$ depends on W_O only through the linear term $\mathrm{Ctx}\, W_O$, whose local Jacobian is Ctx; the chain rule gives $\nabla_{W_O}\mathcal{L} = \mathrm{Ctx}^\top(\partial\mathcal{L}/\partial x_1)$. Both Ctx (cached) and $e_{\mathrm{attn}} = \partial\mathcal{L}/\partial x_1$ (broadcast) are available without transport; the surrogate enters only the *downstream* $\partial\mathcal{L}/\partial\mathrm{Ctx}$, not this term. The MLP case is identical. □

Empirically, the AR-DFA value-path gradients match a causal autograd reference to $\leq 3.1\times10^{-15}$ (Sec. **??**, GUARD 1), confirming both the derivation and the implementation.

4.2 Alignment floor and its decay

Lemma 1 (Depth-dependent alignment floor)**.** *Let ρ_ℓ be the cosine alignment between the transport-free update and the true gradient at layer ℓ. For fixed random broadcast feedback the expected floor scales as $\Theta(1/\sqrt{L})$ in depth L; row-orthonormal (structured) feedback preserves error norm and raises the floor, but does not remove the depth dependence.*

Lemma **??** is the standard structured-feedback intuition made quantitative; crucially, it predicts that alignment *decays* with depth. Our depth study (Sec. **??**) confirms this as a pre-registered negative control: the worst-layer angle of fixed-DFA grows monotonically with depth (slope $+3.26°$/layer, CI excludes 0). This repairs a gap in the earlier version of the theory, which *assumed* a sign-advantage $p > \frac{1}{2}$ and a positive alignment rather than deriving them; here these quantities are *measured*, and Lemma **??**'s sign is validated rather than posited.

4.3 Why Perturbation-Taught feedback lifts the level but not the slope

Proposition 2 (Fixed-linear ceiling of PT)**.** *The expected fixed point of the update* (**??**) *is the linear regression map* $R_O^\star = \mathbb{E}[\hat{g}_{\mathrm{Ctx}}\, e^\top]\,\mathbb{E}[e\, e^\top]^{-1}$*. Consequently* $R_O^\star$ *is a* fixed *linear function of the broadcast error and cannot track an input-conditioned target;* PT *can lower the alignment error* level *but cannot change its scaling with depth. Moreover, the single-sample node-perturbation estimator has variance* $\propto d_{\mathrm{pert}}$*, so reaching a target averaged cosine requires* $N = \Theta(d_{\mathrm{pert}})$ *samples.*

Proposition **??** predicts exactly the empirical pattern below: PT improves mid-depth alignment (a level shift) but leaves the depth-slope unchanged (Sec. **??**), and its surrogate only *partially* recovers $W_O^\top$ at realistic width (relative error 1.06; Sec. **??**). The gain at scale therefore comes from AR-DFA's exact A-reuse, not from PT reaching exactness.

4.4 Stationarity, read honestly

For completeness we record the biased-SGD view. With a β-smooth loss and a transport-free direction $\tilde{g}_t$, the descent lemma gives $\frac{1}{T}\sum_t \mathbb{E}\|\nabla\mathcal{L}(V_t)\|^2 \le \frac{2(\mathcal{L}_0-\mathcal{L}_\star)}{\eta T} + C_1\beta\eta\sigma^2 + C_2(1-2p)^2 + C_0\sum_\ell(1-\bar{\rho}_\ell)$, a standard bound in the sign-advantage p and per-layer alignment $\bar{\rho}_\ell$. We stress what this does and does not say: it is a convergence statement *conditional* on p and $\bar{\rho}_\ell$, which our experiments treat as measured quantities, not free assumptions. It does not bound the *solution quality* relative to BP, and Sec. **??** shows empirically that the achievable stationary point is substantially worse.

5 Experimental Protocol

All experiments are pre-registered with numeric gates and run in float64 where a reference gradient is needed, on CPU with pinned threads and fixed seeds. Autograd is used *only* to produce exact-BP references and alignment targets; the transport-free strategies compute their own gradients explicitly. We report mean ± 95% CIs over $n \ge 3$ seeds. Guards enforced at every stage:

- **GUARD 1 (value-path exactness).** AR-DFA's $\nabla_{W_O}, \nabla_{W_2}$ match a causal autograd reference to $< 10^{-5}$ (measured $\le 3.1 \times 10^{-15}$).
- **GUARD 2 (transport-free).** An AST-level check confirms the surrogate-adaptation path contains no autograd call and no $W_O^\top$; a *positive control* (a variant that does read $W_O^\top$) must be flagged, and an e-fixed weight-perturbation leaves AR-DFA's gradients invariant (change = 0) while changing the positive control ($\approx$ 29–33).
- **Alignment metric.** Per-weight-matrix cosine to the exact reference, reported as the worst (max-angle) matrix; $\theta = \arccos(\rho)$ in degrees; depth-slope by OLS of $\theta_{\min}(L)$.

Artifacts (code, JSON results, seeds) reproduce every table.

6 Results

6.1 Alignment through attention

On a single decoder block ($d = 32$, $n = 5$ seeds), Table **??** reports the worst attention-block alignment angle for the four conditions. AR-DFA paired with PT (AR-DFA+PT) reduces the worst-case angle from 90.6° (fixed-DFA, i.e. random) to 46.4°, a 44.2° gain with non-overlapping CIs. Neither component alone helps: AR-DFA with a *fixed random* surrogate (90.8°) and PT without A-reuse (91.8°) are both at the random floor. The synergy is exactly Proposition **??**/**??**:

the exact A-routing only pays off once PT makes the surrogate non-random (here to relative error 1.06—partial, not exact). The value-path exactness holds throughout (GUARD 1).

condition	worst attention-block θ (deg)	value path	interpretation
fixed-DFA (random)	90.6 ± 0.4	—	baseline
AR-DFA only (random R_O)	90.8 ± 0.9	exact	A-reuse alone insufficient
PT only (no A-reuse)	91.8 ± 1.4	—	surrogate-learning alone insufficient
AR-DFA+PT	$\mathbf{46.4 \pm 5.6}$	exact	$\Delta = 44.2°$, transport-free

Table 1: Attention-block alignment ablation ($d = 32$, $n = 5$). Lower is better; 90° is random. Only the combination of exact A-reuse (AR-DFA) and a learned surrogate (PT) clears the floor.

6.2 Depth: a level shift, not a slope change

On a configurable-depth network (worst-layer angle, $n = 5$ seeds), Table **??** shows PT improving mid-depth alignment—+8.1° at L=4 and +14.7° at L=8—but *not* bending the decay: the depth-slopes are statistically indistinguishable (+3.41 vs +3.26°/layer), and at L=16 both methods have collapsed. The fixed-DFA slope +3.26°/layer (CI ±1.12, excluding 0) is the pre-registered negative control confirming Lemma **??**; PT's behavior is exactly the fixed-linear ceiling of Proposition **??**.

method	L=2	L=4	L=8	L=16	slope (°/layer)
fixed-DFA	55.6	73.9	78.1	105.8	+3.26 (±1.12)
PT (adapted)	56.7	65.7	63.4	106.1	+3.41 (±1.69)

Table 2: Worst-layer alignment angle vs depth. PT lifts the mid-depth *level* but the depth-*slope* is unchanged; both collapse by L=16. Larger angle = worse alignment.

6.3 Accuracy: the alignment win does not transfer

We train a stacked character-level language model on a controlled context-free-grammar corpus (vocabulary 38, bigram floor 2.881 bpc, $n = 5$ seeds) and compare validation bits-per-char. Table **??** is the paper's pivot. AR-DFA+PT is a genuine learner—it beats fixed-DFA by 1.10 bpc and edges below the bigram floor—but it plateaus **1.93 bpc above backpropagation**, a gap $\sim 13\times$ the pre-registered 0.15 bpc gate (CI far above it): a decisive **NO-GO**. The alignment gain of Sec. **??** does not become an accuracy gain.

method	val bpc	vs BP	per-step (rel.)
backpropagation	$\mathbf{0.774 \pm 0.007}$	—	$1\times$
AR-DFA+PT	2.702 ± 0.088	+1.93	$\sim 31\times$
fixed-DFA	3.804 ± 0.055	+3.03	$\sim 11\times$
bigram floor = 2.881 bpc; gate (AR-DFA+PT− BP $\leq$ 0.15, CI upper $<$ 0.25): **FAIL**			

Table 3: Character-level LM (CFG corpus, $n = 5$). AR-DFA+PT beats fixed-DFA and the bigram floor but is 1.93 bpc worse than BP and $\sim 30\times$ slower per step on CPU.

6.4 Economics: throughput is not GPU-dollars

DFA's only genuine efficiency axis is backward *parallelism*, which materializes only in pipeline-parallel training. An analytical cost model (forward t_f=1, backward t_b=2; AR-DFA-pipeline vs tuned 1F1B+checkpointing) gives throughput ratios of 1.15× when the per-stage update cost $t_u \approx t_b$, 1.66× when $t_u \approx t_f$, and 2.2–2.85× when memory pressure forces BP (but not the $O(1)$-activation AR-DFA) to recompute. So the pipeline win is real but *conditional* on the unmeasured t_u, and the model omits the e-broadcast communication, which erodes it further.

This throughput number, however, does not save compute, because the relevant quantity is **wall-clock to a target loss** = (steps-to-target) × (time-per-step). Sec. **??** shows AR-DFA converges to a loss BP surpasses, so time-to-BP-quality is unbounded regardless of throughput. Finally, a transport-fraction frontier—train with update $(1-\lambda)\,\mathrm{DFA} + \lambda\,\mathrm{BP}$—shows accuracy recovers roughly *in proportion* to λ: 25% of the gap at λ=0.10 and 52% at λ=0.25. There is no cheap-closure regime; buying accuracy costs transport, which is precisely the parallelism one set out to save.

7 FeedFlip: Feedback-Driven Bit-Flip Training

We now use the feedback signal in the setting that names the project: training ternary networks by flipping bits directly, with no float shadow weights.

The mechanism. Each weight is stored only as a ternary value $w \in \{-1, 0, +1\}$ and a small signed integer accumulator c. At each step we form a per-weight descent vote from the *sign* of the gradient estimate and integrate it, flipping the bit when the evidence crosses a threshold K:

$$c \leftarrow c - \mathrm{sign}(\hat{g}), \qquad \text{if } c \geq K: \ w \leftarrow \min(w+1, +1),\ c \leftarrow 0; \qquad \text{if } c \leq -K: \ w \leftarrow \max(w-1, -1),\ c \leftarrow 0. \tag{6}$$

The forward pass uses w with a fixed per-layer scale. There is no float shadow and no full gradient—only a *sign*. State per weight collapses from a 32-bit shadow to $\approx 1.58 + \log_2(2K+1)$ bits (e.g. 5.7 bits at K=8). Because a flip needs only the sign, DFA is the natural feedback: it supplies that sign transport-free, shadow-free, and lock-free.

The mechanism works—with a good sign. On a deep MLP (teacher task, n=3; Table **??**), sign-driven flipping with the *exact* (BP) sign matches float-shadow BP (0.629 vs 0.619) at 5.6× less state and ≈2× faster steps. The bit-flip idea is real. Transport-free flipping is partially viable here: vote-averaging (larger K) plus orthogonal feedback compound to 0.548—within 0.08 of BP, transport-free, at ≈4× less state—but only partially.

sign source (MLP)	final acc	ms/step	state bits/w
float-shadow BP	0.619	2.06	32.0
flip, BP sign (K=8)	**0.629**	1.03	**5.67**
flip, DFA sign (K=8)	0.313	0.81	5.67
flip, DFA+ortho (K=32)	0.548	0.81	7.60

Table 4: FeedFlip on a deep MLP. Sign-flips with an accurate (BP) sign match float-shadow BP at ≈5.6× less state; transport-free flips are only partially viable (best 0.548).

On a Transformer, transport-free flipping fails—and AR-DFA does not rescue it. We repeat the experiment on a stacked ternary character-level Transformer LM (CFG corpus, floor 2.881 bpc, n=3; Table **??**). With the exact sign, FeedFlip works (1.72 bpc, past the floor and the FROZEN control). But *every* transport-free arm is **worse than FROZEN** (blocks never flip): flipping bits from the DFA sign is worse than not flipping at all, and vote-averaging (K=32) only claws back to the floor. Crucially, AR-DFA—with its far better cosine alignment—does *not* beat vanilla fixed-DFA.

arm (ternary Transformer)	val bpc	vs floor	bits/w	sign-match p
float-shadow BP	**1.12**	−1.76	32.0	—
flip, BP sign	**1.72**	−1.16	5.67	—
FROZEN (no flips)	2.01	−0.87	5.67	—
flip, AR-DFA (K=8)	3.28	+0.40	5.67	≈ 0.53
flip, AR-DFA (K=32)	2.80	−0.08	7.60	—
flip, fixed-DFA (K=8)	3.21	+0.33	5.67	≈ 0.53
flip, fixed-DFA (K=32)	2.95	+0.07	7.60	—

Table 5: FeedFlip on a ternary Transformer. Transport-free arms are all *worse than FROZEN*; AR-DFA ties fixed-DFA. Per-weight sign-match $p\approx 0.53$ (near the 0.5 chance line) for both.

Why: cosine alignment is not sign correctness. The mechanistic explanation is the last column of Table **??**. A bit-flip consumes only the per-weight *sign* of the gradient, and the fraction of weights whose transport-free sign matches the true gradient is $p\approx 0.53$ for *both* fixed-DFA and AR-DFA—barely above the 0.5 chance line—*even though* AR-DFA improves the aggregate *cosine* alignment from 90° to 46° (Table **??**). A large cosine gain can come almost entirely from a few high-magnitude coordinates and a global rotation while leaving most per-coordinate signs unchanged; the flip rule, which weights every coordinate equally by sign, is blind to exactly the improvement AR-DFA provides. **Cosine alignment—the standard DFA diagnostic—is the wrong proxy for discrete, transport-free learning; per-weight sign correctness is what a bit-flip needs, and it is the quantity that does not improve.**

8 The Sign Barrier

The flip experiments and the continuous-training experiments (Secs. **??**–**??**) tell one story from two directions.

- **The FeedFlip mechanism is a genuine efficiency win**—shadow-free, sign-driven, $\approx 5.6\times$ less state, $\approx 2\times$ faster—*given* an accurate sign.
- **Transport-free feedback supplies direction, not per-weight sign.** AR-DFA maximizes the aggregate (cosine) direction achievable transport-free, yet its per-weight sign-match stays near chance. The discrete regime, which uses only signs, therefore gains nothing; and the continuous regime, which can exploit magnitude, still does not reach competitive accuracy, per-step cost, or wall-clock-to-target.
- **The two failures share one cause.** The transport-fraction frontier shows continuous accuracy recovers only in proportion to reintroduced transport, and the flip study shows the sign it needs is not there transport-free. Both are the same wall: closing the gap—in signs or in magnitudes—demands the transport DFA removes, and the methods that supply it (predictive coding, target propagation) re-lock the backward.

The practical implication is a redirection with a sharper target. Effort spent improving *cosine* alignment (the usual objective of better-feedback methods) has diminishing returns for either regime; the open problem is a transport-free rule whose *per-weight sign* is reliably correct at depth and through attention. Our evidence suggests this is hard, and—given that the sign barrier and the accuracy wall coincide—possibly the same problem.

9 Limitations

Our study is deliberately laptop-scale and CPU-only: single-head blocks, small corpora, and modest depth, chosen so that every claim is exactly reproducible and the reference gradients are affordable. The FeedFlip experiments share this scope (deep MLP and a small ternary Transformer, n=3), and the sign-match p is measured over nonzero-true-gradient entries at a few training snapshots; the efficiency claim for the mechanism is established with an *accurate* sign at this scale, not at LM scale. The pipeline cost model is analytical and not yet validated against a multi-process simulator or GPU hardware; in particular t_u (the per-stage update cost) is the pivotal unmeasured quantity, and the e-broadcast communication is omitted. The transport-fraction frontier is measured on a task with a modest DFA–BP gap and is suggestive rather than decisive at the large-gap LM scale. These bound the strength of the positive claim (a real but small-scale mechanism) more than the negative one, which is corroborated across independent axes (discrete flips and continuous training).

10 Conclusion

FeedFlipNets trains ternary networks by feedback-driven bit-flips: no float shadow weights, a per-weight sign vote flips a bit when evidence accrues, and—because a flip needs only the sign—DFA supplies that signal transport-free and shadow-free. The mechanism is genuinely efficient given an accurate sign ($\approx$5.6$\times$ less state, $\approx$2$\times$ faster than float-shadow BP). To provide a better transport-free sign we introduced Activation-Routed DFA, which reuses a Transformer block's cached data-maps and surrogates only the weight transposes, yielding exact value-path gradients and cutting the worst-case attention *cosine*-alignment angle from 90° to 46°. The central finding is negative and precise: this alignment gain does not help. On a ternary Transformer, transport-free flipping is worse than not flipping, and AR-DFA does not beat vanilla DFA, because per-weight *sign*-match stays near chance even as cosine alignment improves sharply—and the same barrier shows up on the continuous side as an unclosed accuracy and wall-clock-to-target gap. The lasting message: cosine alignment is the wrong objective for transport-free learning; per-weight sign correctness is the binding constraint, in the discrete regime directly and in the continuous regime through the transport-fraction frontier. We hope the efficient bit-flip mechanism, and especially the honestly-negative sign-barrier characterization, help others target the right quantity.

A Ternary-forward stability baseline (secondary)

An earlier version of this project studied deterministic DFA with ternary forward weights $W = Q_\tau(V)$ (float "shadow" weights V updated by DFA, ternary W periodically "flipped" from V) as a stability-and-convergence baseline on MNIST, Fashion-MNIST, AG News, Adult, California Housing, 20 Newsgroups, and UCR GunPoint. That study targets deterministic, reproducible baselines rather than state-of-the-art accuracy and is orthogonal to the mechanism above; its sweeps (learning rate, threshold τ, flip schedule), alignment/stability diagnostics, and the corresponding tables and figures are retained in the project artifacts. The present paper's contribution—AR-DFA, its value-path exactness, and the alignment–accuracy characterization—is in full precision and does not depend on ternary quantization.

B Reproducibility

Each result section corresponds to a pre-registered gate with a released JSON artifact: attention ablation (Table **??**), depth sweep (Table **??**), LM bits-per-char (Table **??**), and the cost model / transport frontier (Sec. **??**). Guards (value-path exactness, transport-free with positive control, lock-free e-fixed probe) are unit-tested. All runs use fixed seeds, pinned BLAS threads, and float64 references; the transport-free strategies never invoke autograd for their own gradients.